

Software Requirements Specification (SRS)

Project: AI Citizen Complaint Intelligence System

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1. Introduction

1.1 Background

Government institutions in Pakistan receive a massive volume of citizen complaints through online portals such as the Pakistan Citizen Portal. Complaints vary in quality, language, urgency, and relevance. Currently, most complaints are handled manually, leading to delays, misclassification, poor prioritization, and unresolved issues. This reduces governance efficiency and public trust.

With AI and NLP advancements, unstructured complaint text can be intelligently analyzed to automate classification, prioritization, routing, and performance tracking.

1.2 Problem Statement (Detailed)

Government complaint management systems face these critical operational problems:

- 1. Unstructured Complaint Text:** *Citizens submit complaints in free-text form, often poorly written or incomplete.*
- 2. Manual Classification & Routing:** *Human error causes delays in assigning complaints to departments.*
- 3. Lack of Urgency Detection:** *Emergency or life-threatening complaints are treated the same as routine complaints.*
- 4. Duplicate & Repeated Complaints:** *Multiple citizens report the same issue, wasting resources.*
- 5. Hidden Corruption Indicators:** *Complaints containing bribery or misuse of authority are not proactively flagged.*

6. **Poor Accountability & Transparency:** Limited tracking of response quality, officer performance, and resolution timelines.
7. **Language Barrier:** Complaints submitted in Urdu are harder to process due to limited NLP support.

1.3 Objectives

Major Objectives:

- Automatically classify citizen complaints into departments.
- Detect complaint urgency and emotional intensity.
- Route complaints to correct departments without human intervention.
- Identify potential corruption-related complaints.
- Provide analytical dashboards for monitoring resolution performance.

Minor Objectives:

- Support bilingual processing (Urdu & English).
- Reduce complaint processing time.
- Minimize duplicate complaints using similarity detection.
- Improve citizen satisfaction through faster responses.
- Create a scalable SaaS architecture for national deployment.

2. Solution Statement

The AI Citizen Complaint Intelligence System is a **web-based platform** using AI/NLP models to analyze, classify, prioritize, and manage complaints automatically. It acts as an **AI decision-support layer**, enabling:

- Faster resolution
- Accurate classification & prioritization

- *Corruption detection*
- *Officer accountability*
- *Policy insights for government*

3. System Scope

- *Complaint intake via web & mobile portals*
- *Automatic language detection (Urdu/English)*
- *AI-based department classification*
- *Urgency & sentiment scoring*
- *Duplicate complaint clustering*
- *Corruption risk identification*
- *Automated routing & escalation*
- *Performance tracking dashboards*
- *Secure access control*
- *Scalable deployment for national use*

4. Overall System Description

4.1 System Perspective

Three-tier architecture:

1. **Presentation Layer (Frontend):** *React.js dashboards, citizen & officer interfaces*
2. **Application Layer (Backend & AI Engine):** *Python/Django, NLP models, routing & analytics*
3. **Data Layer (Databases & Logs):** *MongoDB/PostgreSQL*

4.2 User Classes & Characteristics

User	Description	Skill	Responsibilities
Citizen	Submits complaints	Low	Complaint submission, track status
Officer	Handles complaints	Medium	Respond & resolve complaints
Department Admin	Supervises officers	High	Monitoring & analytics
System Admin	System management	High	User management, logs, system health

5. Features (High-Level)

- Complaint Classification
- Urgency Detection
- Corruption Risk Detection
- Duplicate Complaint Detection
- Smart Routing
- Performance Analytics
- Secure Role-Based Access
- AI-Assisted Suggested Solution

6. Functional Requirements (Detailed)

6.1 Complaint Submission

- Web-based complaint form

- *Field validation*
- *Support for Urdu & English input*
- *Store complaints in raw & processed form*

6.2 Text Preprocessing

- *Tokenization, stop-word removal*
- *Noise cleaning & normalization*
- *Urdu text normalization*

6.3 Complaint Classification

- *Department & category assignment*
- *Confidence scoring*

6.4 Urgency & Sentiment Detection

- *Emotional tone detection*
- *Urgency scoring*
- *Automatic escalation if threshold exceeded*

6.5 Duplicate Detection

- *Semantic similarity comparison*
- *Complaint clustering*
- *Flagging duplicates*

6.6 Corruption Risk Detection

- *Identify bribery/misuse keywords*
- *Risk scoring*

6.7 Routing & Escalation

- *Automated routing to departments*
- *Officer notifications via email/SMS*
- *Escalation for unresolved urgent complaints*

6.8 Dashboard & Reporting

- *Real-time complaint tracking*
- *Departmental & officer analytics*
- *Exportable reports (CSV/PDF)*

6.9 Security

- *Authentication & authorization*
 - *Role-based access*
 - *Audit logs*
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7. Non-Functional Requirements

7.1 Performance

- *Classification response < 5 sec*
- *Handle 50,000 concurrent complaints*

7.2 Scalability

- *Horizontal scaling*
- *Multi-tenant SaaS support*

7.3 Reliability

- *99.9% uptime*
- *Automatic backups*

7.4 Usability

- *Simple UI for non-technical users*
- *Urdu language support*

7.5 Security

- *Data encryption at rest & in transit*
- *Compliance with data protection laws*

7.6 Maintainability

- *Modular codebase*
- *AI model retraining support*
- *Easy updates*

8. Workflow (Minor + Major Steps)

1. *Complaint submission*
2. *Data validation*

3. *Language detection*
 4. *NLP preprocessing*
 5. *Classification*
 6. *Urgency & sentiment scoring*
 7. *Duplicate detection*
 8. *Corruption risk analysis*
 9. *Routing & escalation*
 10. *Dashboard monitoring & feedback*
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9. Resources Required

Human Resources

- *Data Scientist / AI Lead*
- *Backend Developer*
- *Frontend & Documentation Lead*

Technical Resources

- *GPU-enabled machine*
 - *Cloud hosting (AWS/Azure/GCP)*
 - *NLP libraries (Transformers, UrduHack)*
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10. Pros & Cons

Pros:

- *Solves a real-world governance problem*
- *Strong AI/NLP application*
- *High academic & commercial value*

Cons:

- *Urdu NLP complexity*
 - *Data availability challenges*
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11. Technology Stack

11.1 Frontend

- *React.js, HTML5, CSS3*

11.2 Backend

- *Python (Django/Flask), REST APIs*

11.3 Database

- *MongoDB, PostgreSQL*

11.4 AI / NLP

- *BERT/XLM-R, UrduHack*
- *Sentiment, urgency detection*
- *Duplicate detection using embeddings*

11.5 Deployment

- *Google Colab for model training*
 - *AWS / Azure for cloud deployment*
 - *Docker for containerization*
 - *GitHub for version control*
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12. Work Division (3 Members)

Member	Responsibilities
<i>Data Scientist / AI Lead</i>	<i>Dataset labeling, preprocessing, NLP & ML modeling, duplicate & corruption detection</i>
<i>Backend Developer</i>	<i>API development, workflow logic, AI integration, routing, DB design</i>
<i>Frontend & Documentation Lead</i>	<i>UI dashboards, visualizations, multilingual support, reports, SRS, defense prep</i>

13. Timeline (Two Semesters)

Semester 1 – FYP-1 (Planning & Design)

- *Weeks 1–2: Proposal & problem finalization*
- *Weeks 3–4: Literature review*

- *Weeks 5–6: Existing system analysis*
- *Weeks 7–8: SRS documentation*
- *Weeks 9–10: Dataset collection*
- *Weeks 11–12: Feature engineering plan*
- *Weeks 13–14: System architecture design*
- *Weeks 15–16: FYP-1 Defense*

Semester 2 – FYP-2 (Implementation & Evaluation)

- *Weeks 1–4: NLP preprocessing*
- *Weeks 5–6: Model training*
- *Weeks 7–8: Backend integration*
- *Weeks 9–10: UI dashboards*
- **Mid Defense:** *Working prototype*
- *Weeks 11–13: Optimization & testing*
- *Weeks 14–15: Performance evaluation*
- *Week 16: Final report & demo*

14. Existing Solutions & Gap Analysis

Existing System

*Pakistan Citizen
Portal*

Limitations

Manual routing, no AI

*Local Govt Complaint
Cells*

*Spreadsheet-based, no
analytics*

*Basic Ticketing
Systems*

No NLP intelligence

Gaps:

- *No urgency detection*
- *No corruption flagging*
- *No duplicate clustering*
- *No Urdu NLP*
- *No officer analytics*

15. Unique Factors (6 Key Innovations)

1. ***Urdu + English AI intelligence***
2. ***Corruption risk scoring***
3. ***Urgency-aware routing***
4. ***Duplicate complaint clustering***
5. ***Officer performance analytics***
6. ***Governance intelligence dashboard with policy insights***

17. Users & Interfaces

<i>User</i>	<i>Interface</i>	<i>Login</i>
<i>Citizen</i>	<i>Complaint form, status tracker</i>	<i>Optional / verification code</i>
<i>Officer</i>	<i>Dashboard: problem + AI solution + metrics</i>	<i>Required</i>
<i>Department Admin</i>	<i>Summary dashboard, performance analytics</i>	<i>Required</i>
<i>System Admin</i>	<i>User management, logs, AI retraining</i>	<i>Required</i>